

Lecture 05 - Distributions and the Normal Model

Lecture 4 introduced random variables, expected values, and variances. Lecture 5 moves from general random variables to probability distributions, with special attention to continuous random variables and the normal model. The central change is conceptual: for a discrete random variable, probability can be assigned to exact values; for a continuous random variable, probability is assigned to intervals.

This distinction matters in finance and business because many important variables are treated as continuous: returns, prices after transformation, waiting times, demand, delivery time, revenue growth, loss severity, and measurement error. We rarely ask whether a continuous variable is exactly equal to one number. We ask whether it is below a threshold, above a threshold, or inside an interval.

The lecture has two connected parts. The first part introduces continuous random variables and probability density functions. It explains why probability is area under a curve, not the height of the curve itself. The second part focuses on the normal model: how its mean and standard deviation shape the curve, how probabilities are computed with the cumulative distribution function, how standardization works, and why empirical data must be checked before a normal model is trusted.

1. From Discrete Probabilities to Continuous Probabilities

In a discrete probability model, the random variable can take a finite or countable set of values. For example, the number of defaults in a small portfolio might be 0, 1, 2, or 3. A probability distribution can then be written as a table:

Number of defaults	Probability
0	0.70
1	0.20
2	0.08
3	0.02

The probability of one exact value is meaningful. We can say $P(X = 1) = 0.20$, and that statement has a direct interpretation.

For a continuous random variable, the situation is different. Suppose X is a waiting time, a demand level, or a return. The possible values are no longer a short list. Between 0 and 10 there are infinitely many possible numbers. In a continuous model, probability is not assigned to one exact point. Instead, probability is assigned to an interval such as

$$P(a < X < b).$$

This is not a technical detail. It changes the way we read and interpret probability. A continuous model does not usually answer “What is the probability that X is exactly 2.000000?” It answers questions such as “What is the probability that delivery time is at most 2 days?” or “What is the probability that a daily return is below -3 percent?”

2. Continuous Random Variables

A continuous random variable is a random variable whose possible values lie on an interval or a continuum. It can take any value between two endpoints, or in some cases any value above or below a threshold.

If X is uniformly distributed between 0 and 10, every subinterval of the same length has the same probability. The probability that X falls between a and b , where $0 \leq a < b \leq 10$, is

$$P(a < X < b) = \frac{b - a}{10}.$$

The numerator is the length of the interval of interest. The denominator is the total length of the possible range. This simple example gives the first intuition for continuous probability: probability depends on the size of the interval, not on an isolated point.

In a continuous model,

$$P(X = x) = 0$$

for any exact value x . This does not mean that x is impossible in an ordinary language sense. It means that one point has no width, and therefore no area. Since continuous probabilities are areas, a point has probability zero.

3. Probability Density Functions

A probability density function, or PDF, is a curve that describes how probability is distributed across the possible values of a continuous random variable.

For a function $f(x)$ to be a valid density, it must satisfy two conditions.

First, density cannot be negative:

$$f(x) \geq 0$$

for every possible value of x .

Second, the total area under the density over the full range of possible values must equal 1:

$$\int f(x) dx = 1.$$

The height of the curve is density. The area under the curve is probability. This is the most important sentence in the first part of the lecture. A high density means that values near that part of the axis are relatively concentrated. But probability itself is obtained by taking the area under the curve across an interval.

Figure 1. For a continuous random variable, $P(a < X < b)$ is the shaded area under the density curve between a and b .

4. Describing a Continuous Random Variable

For a discrete random variable, describing the distribution means listing all possible values and their probabilities. For a continuous random variable, that strategy does not work because there are infinitely many possible values.

Instead, we describe two things:

1. the range of possible values;
2. the probability density function over that range.

The curve is the model. It tells us where probability is concentrated and where values are less likely. Equal-width intervals under higher parts of the curve receive more probability than equal-width intervals under lower parts of the curve.

This is why visual interpretation is important. A density curve can be symmetric or skewed, concentrated or dispersed, single-peaked or multi-peaked, bounded or unbounded. Those features matter because they affect the probabilities we compute and the decisions we make.

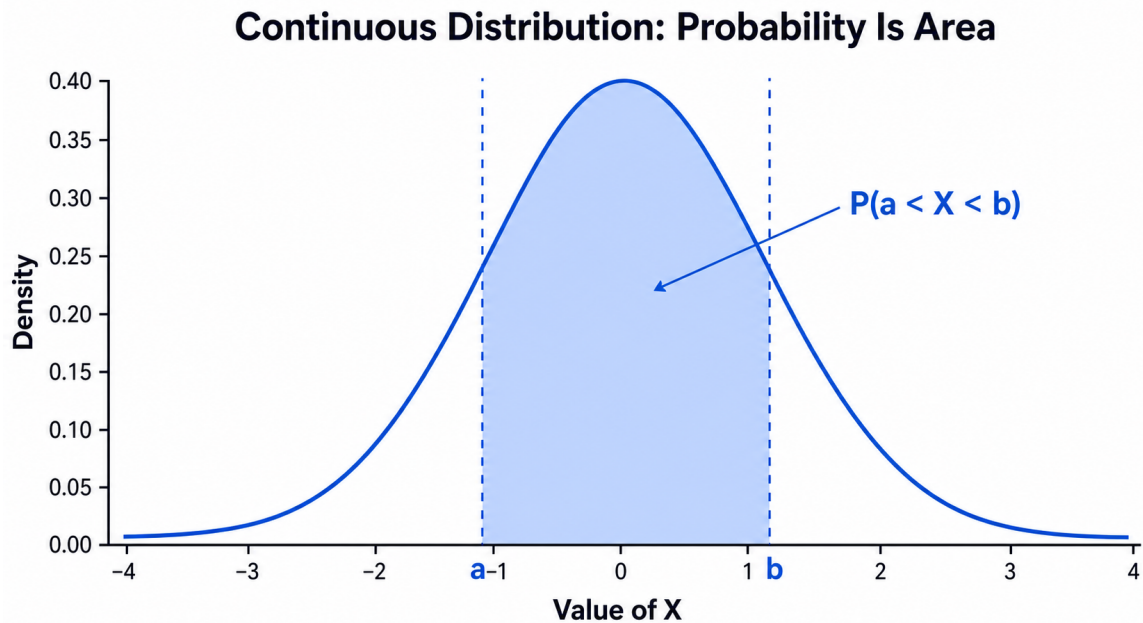


Figure 1: Area under a continuous density curve

5. The Uniform Distribution

The uniform distribution is the simplest continuous distribution. It describes a random variable whose values are spread evenly across an interval.

If

$$X \sim U(a, b),$$

then X can take values between a and b , and the density is constant across that interval:

$$f(x) = \frac{1}{b - a}$$

for $a \leq x \leq b$, and $f(x) = 0$ outside the interval.

The height $1/(b - a)$ is chosen so that the total area is 1. The base of the rectangle is $b - a$, and the height is $1/(b - a)$, so the area is

$$(b - a) \times \frac{1}{b - a} = 1.$$

Under a uniform distribution, probability over a subinterval depends only on the length of the subinterval. If two intervals have the same length, they have the same probability.

6. The Exponential Distribution

The exponential distribution is often used to model waiting times. Examples include the time until the next customer arrives, the time until a machine fails, the time between orders, or the time until a service request is received.

An exponential density has the form

$$f(x) = \lambda e^{-\lambda x}$$

for $x \geq 0$, where $\lambda > 0$ is the event rate.

The restriction $x \geq 0$ is important. Waiting time cannot be negative. The density is highest near zero and then declines as x increases, meaning that short waiting times are more likely than long waiting times.

The parameter λ controls the speed of the process. A large λ means events happen frequently, so the density is concentrated near zero. A small λ means events happen less frequently, so longer waiting times become more plausible.

The exponential distribution is not symmetric. It is right-skewed. That makes it very different from the normal distribution and reminds us that not every continuous variable should be modeled with a bell curve.

7. The Normal Distribution

The normal distribution is the most widely used continuous distribution in introductory statistics. It describes variables that are concentrated around a central value, with symmetric variation on both sides.

We write

$$X \sim N(\mu, \sigma^2)$$

or, in some software-oriented notation,

$$X \sim N(\mu, \sigma),$$

where μ is the mean and σ is the standard deviation. The mean determines the center of the distribution. The standard deviation determines the spread.

The density is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right].$$

The formula looks more complicated than the idea. Values close to μ have high density. Values far from μ have low density. The curve is symmetric around the mean, and the total area under it is 1.

The normal model is often used for measurement errors, demand fluctuations, average effects of many small independent shocks, and approximate descriptions of returns near the center of their distribution. It is powerful, but it is still a model. It must be checked against the data.

8. Mean and Standard Deviation in the Normal Model

The two parameters of the normal model have distinct roles.

The mean μ shifts the curve left or right. If μ increases, the entire distribution moves to the right. If μ decreases, the entire distribution moves to the left.

The standard deviation σ changes the spread. A small σ gives a narrow and tall curve because values are tightly concentrated around the mean. A large σ gives a wider and lower curve because probability is spread over a larger range.

This link between σ and the curve height is not arbitrary. The total area under the curve must always remain 1. If the distribution becomes wider, it must also become lower; otherwise the total area would exceed 1.

9. Probabilities Under the Normal Curve

For a normal random variable, probabilities are areas under the normal density curve.

For example,

$$P(a < X < b)$$

is the area under the normal curve between a and b . Unlike the uniform distribution, this area is not a rectangle. There is no simple elementary formula for the area between two cutoffs. That is why normal probabilities are usually computed with software or probability tables.

The geometric interpretation remains essential. If we ask for

$$P(X < 120),$$

we are asking for all the area to the left of 120. If we ask for

$$P(90 < X < 120),$$

we are asking for the area between 90 and 120.

10. The Cumulative Distribution Function

The cumulative distribution function, or CDF, is the practical tool used to compute probabilities.

For a random variable X , the CDF is

$$F(x) = P(X \leq x).$$

It gives the probability accumulated from the far left of the distribution up to x .

Once we have the CDF, an interval probability can be computed by subtraction:

$$P(a < X < b) = F(b) - F(a).$$

This formula says: take all the probability to the left of b , subtract all the probability to the left of a , and what remains is the probability between a and b .

In Python, normal CDF values are usually computed with a function such as `scipy.stats.norm.cdf`. Conceptually, however, the key point is not the code. The key point is that interval probabilities are differences between cumulative probabilities.

11. Applied Normal Probability Example

Suppose daily product demand is modeled as

$$X \sim N(100, 15^2).$$

The mean demand is 100 units, and the standard deviation is 15 units. We want to compute

$$P(90 < X < 120).$$

Using the CDF,

$$P(90 < X < 120) = F(120) - F(90).$$

With a normal calculator or Python, this probability is approximately

$$0.6563.$$

Under this normal model, the probability that daily demand falls between 90 and 120 units is about 65.6 percent.

Notice the wording. This is not a guarantee about tomorrow. It is a probability under the assumed normal model. If the model is not appropriate for demand, the probability may be misleading.

12. Standardization and Z-Scores

Standardization converts a normal random variable into a standard normal random variable.

If

$$X \sim N(\mu, \sigma^2),$$

then

$$Z = \frac{X - \mu}{\sigma}$$

follows a standard normal distribution:

$$Z \sim N(0, 1).$$

The subtraction $X - \mu$ recenters the variable so that the mean becomes zero. The division by σ rescales the variable so that one unit on the new scale equals one standard deviation.

A z-score tells us how many standard deviations a value is above or below the mean:

$$z = \frac{x - \mu}{\sigma}.$$

For the demand example with $\mu = 100$ and $\sigma = 15$, the value $x = 120$ has z-score

$$z = \frac{120 - 100}{15} = 1.33.$$

So 120 units is 1.33 standard deviations above the mean. A positive z-score is above the mean; a negative z-score is below the mean; a larger absolute z-score is farther from the center.

Standardization remains useful even though software can compute probabilities directly. It gives a universal scale for comparing values measured in different units.

13. From Data to a Fitted Distribution

The lecture then moves from theoretical distributions to empirical data.

An empirical distribution describes the observed sample. A histogram of observed returns, waiting times, or customer spending is an empirical object. It summarizes what happened in the data.

A fitted distribution is a theoretical model chosen to approximate that empirical shape. For a normal model, the fitted parameters are usually

$$\hat{\mu} = \bar{x}$$

and

$$\hat{\sigma} = s,$$

where $\bar{x}$ is the sample mean and s is the sample standard deviation.

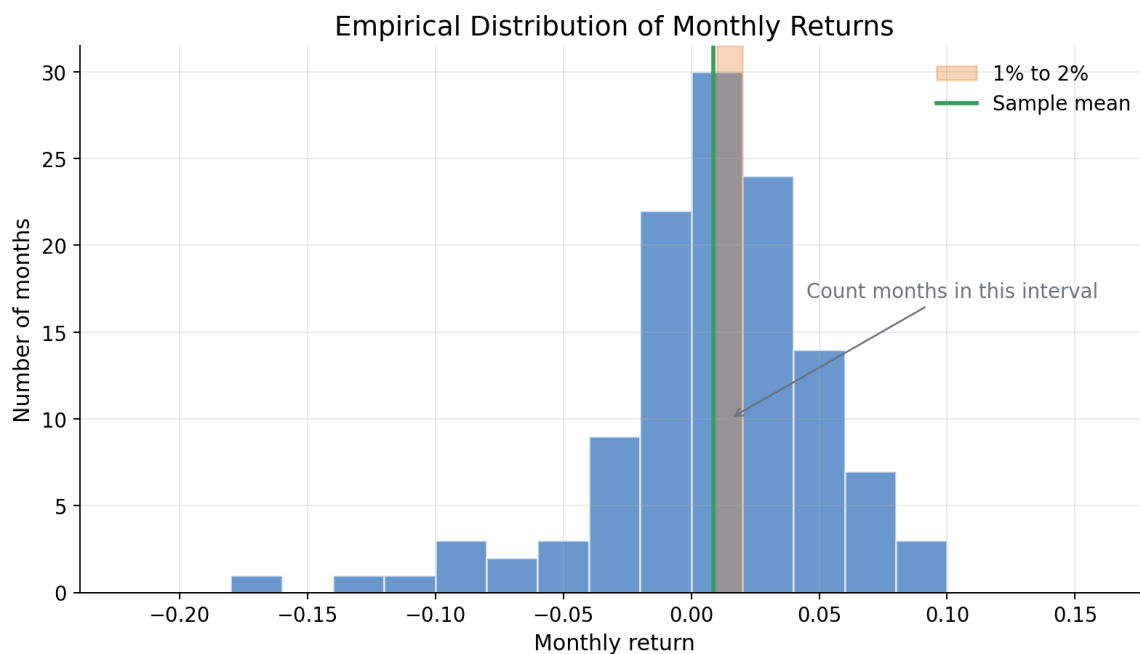


Figure 2: Empirical distribution of monthly returns

Figure 2. A histogram is an empirical distribution: it describes the observed sample, not an assumed probability model.

Fitting a distribution is only the first step. A model can be convenient and still be wrong in important places. After fitting, we must compare the model to the data.

14. Fitting a Normal Model to Netflix Log Returns

The lecture uses Netflix daily log returns from 2015 to 2026 as a real-data example. The variable of interest is the daily log return. The workflow is:

1. read the return data;
2. compute the sample mean and sample standard deviation;
3. fit a normal distribution using those two parameters;
4. compare the empirical histogram with the fitted normal density;
5. compare empirical probabilities with probabilities implied by the normal model.

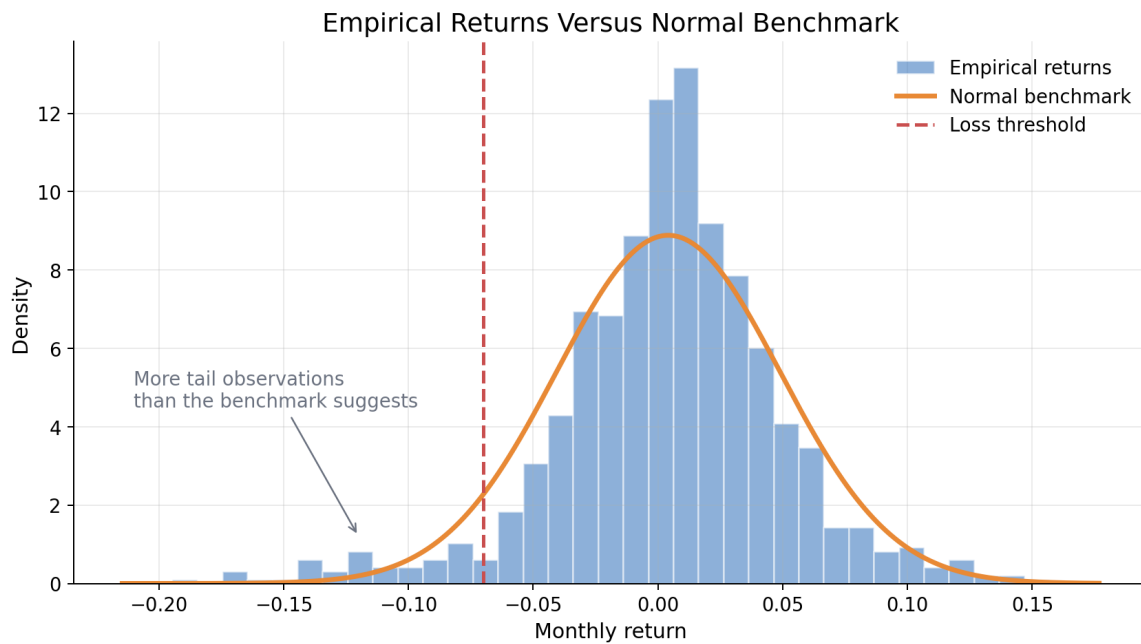


Figure 3: Empirical returns versus a fitted normal benchmark

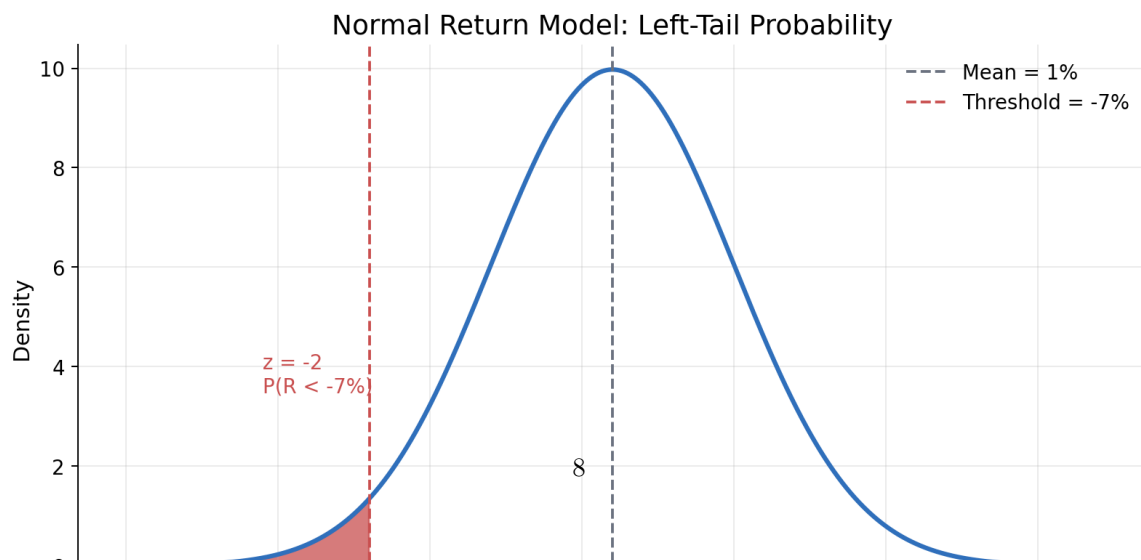
Figure 3. The fitted normal curve can approximate the center of a return distribution while missing important tail behavior.

For the Netflix data used in the slides, the sample mean daily log return is about 0.109 percent and the sample standard deviation is about 2.893 percent. The fitted normal model is therefore centered close to zero but allows daily movements of several percentage points.

The key question is not only whether the fitted curve looks plausible. The key question is whether the model gives reasonable probabilities for the events we care about.

Probability question	Empirical data	Normal model
$P(-0.02 < X < 0.02)$	68.0%	51.0%
$P(-0.05 < X < -0.02)$	11.5%	19.4%
$P(X < -0.03)$	8.5%	14.1%
$P(X > 0.04)$	5.4%	8.9%
$P(X < \mu - 3\sigma)$	0.70%	0.135%

The table shows that the normal model is not uniformly wrong in one direction. In some regions it puts too much probability; in others it puts too little. The most important warning is the final row: extremely negative returns occur more often in the empirical data than the fitted normal model predicts.



It can be problematic when the data are:

1. strongly skewed;
2. bounded below by zero;
3. dominated by extreme values;
4. clearly multi-modal;
5. affected by rare but large shocks.

Waiting times, for example, cannot be negative and are often right-skewed. Customer spending may have a long right tail. Financial returns may have heavy tails, especially on the downside. In all of these cases, a normal model may hide features that matter.

The correct habit is not to reject the normal model automatically. The correct habit is to use it as a benchmark, then check it. Compare the model with the histogram. Compare theoretical probabilities with empirical frequencies. Pay special attention to the events that matter for the decision.

16. Careful Interpretation

A probability computed from a model should always be interpreted as a model-based statement.

For example, the sentence

The probability of a daily Netflix log return below -3 percent is 14.1 percent.

is incomplete unless it is clear that this is under the fitted normal model. A more careful sentence is:

Under the fitted normal model, the probability of a daily Netflix log return below -3 percent is about 14.1 percent; in the observed sample, the empirical frequency is about 8.5 percent.

The second sentence separates the model from the data. That distinction is essential. Empirical frequencies describe the observed sample. Model probabilities describe an assumed probability model. Both can be useful, but they are not the same object.

This lecture prepares the ground for sampling distributions and confidence intervals. Once we can describe distributions and fit models, the next question is how estimates vary from sample to sample and how precise our estimates are likely to be.